INCH-POUND
MIL-DTL-24643/20G
27 November 2023
SUPERSEDING
MIL-DTL-24643/20F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LS7SGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Thermoset insulation (see table I for thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering (unless required by 3.3.6 of MIL-DTL-24643).
Fifth	-	The seven conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Standard identification code applied by method 1. Outer surface to be printed shall appear black.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/20	Type and size	Conductor size (AWG)	Cable overall diameter max. (inch)	Conductor resistance max. (ohms)	Insulation thickness min. avg. (inch)
-01UN	LS7SGU-3	16 (class B)	0.545	4.3	0.018
-02UN	LS7SGU-4	14 (class B)	0.595	2.68	0.018
-03UN	LS7SGU-6	12 (class B)	0.630	1.72	0.018

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Abrasion resistance (extruded insulation only) (scrapes, min.)	250
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance (extruded insulation only)	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	

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Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS7SGU all sizes	2,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-036)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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